

Space heating and domestic hot water provision for multi-apartment buildings, light commercial premises, schools, and retrofit social housing projects requiring a single packaged unit with low environmental impact.

Fractional Forge — Project 30kw_heat_pump_r290_4500_gbp

Generated: 2026-05-05

Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

Not computed

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed

[B] Research

SUBSYSTEM MODULES

0

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source
- [D] Algorithmic estimate or LLM extraction
- [E] LLM assumption or unverified hypothesis

Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

Check	Status	Reason	Evidence
1. Market TAM/SAM/SOM Defined	FAIL	Missing market data	Not computed
2. Regulatory Standards Identified	PASS	Standards found	6 standards
3. Cost Ceiling Specified	WARN	Unconstrained economics	Not computed
4. Spatial Envelope Provided	WARN	No dimensions	Not computed
5. Modules Decomposed	FAIL	Decomposition failed	0 modules
6. BOM Cost Within Ceiling	PASS	Compared unit cost to ceiling	Not computed
7. Risk Matrix Saturated	FAIL	FMEA rows generated	Risks exist

Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

Data Sources

[System] Internal gate checks

Overall Section Grade: A

1. Brief & Requirements [A]

1.1 Mission & Use Case

Mission

To deliver a high-efficiency, low-GWP 30kW R290 heat pump at an accessible price point that accelerates commercial and large residential decarbonisation.

Use Case

Space heating and domestic hot water provision for multi-apartment buildings, light commercial premises, schools, and retrofit social housing projects requiring a single packaged unit with low environmental impact.

1.2 Market Context & Timing [B]

The UK Boiler Upgrade Scheme, rising gas prices, EU and UK F-Gas refrigerant phase-down regulations, and installer market maturity are converging to create strong demand for affordable, natural refrigerant heat pumps in the 20–50kW range.

Social housing developers, light commercial building owners, multi-family residential landlords, M&E contractors, HVAC system integrators, and local authorities pursuing Net Zero retrofits

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

1.3 Competitor Landscape [C]

Viessmann	
Product	Vitocal 200-A / 250-A AWO series (R290, up to 60kW)
Pricing	Estimated 9,000–14,000 GBP for 30kW equivalent configuration (trade price)
Tech Specs	R290 refrigerant, modulating inverter compressor, SCOP up to 5.0 (A7/W35), operating range -20°C to +35°C ambient, low-noise scroll compressor, integrated hydraulic module available, weather-compensated controls
STRENGTHS Strong brand recognition among UK installers, extensive dealer network, high SCOP performance, full MCS certification, comprehensive warranty and aftermarket support WEAKNESSES High price point significantly above 4,500 GBP target, complex ordering and configuration process, lead times affected by European supply chain, not designed for low-cost market segment	
DIFFERENTIATION ANGLE A 4,500 GBP R290 unit at equivalent performance undercuts Viessmann by 50–70%, making it accessible to social housing and budget-constrained commercial projects that cannot justify premium pricing	

Stiebel Eltron

Product	WPL-A 30 Premium (R290 air source, 30kW nominal)
Pricing	Approximately 8,500–12,000 GBP trade price for 30kW configuration
Tech Specs	R290 refrigerant, 30kW heating capacity at A2/W35, inverter-driven rotary compressor, SCOP 4.47 (EN14825 average climate), operating range -25°C to +40°C, integrated controller, weight approx 280kg, sound power level 68 dB(A)

STRENGTHS

Established R290 expertise, German engineering reputation, full EU and UK certification, good installer training programme, strong SCOP figures, available through major UK distributors

WEAKNESSES

Premium pricing excludes cost-sensitive segments, limited flexibility for OEM or system integrator customisation, noise levels may be challenging for dense urban residential settings

DIFFERENTIATION ANGLE

Price positioning at 4,500 GBP is approximately 50% below Stiebel Eltron, enabling market penetration in local authority, housing association, and light commercial segments where capital cost is the primary purchase driver

Midea / Carrier (OEM)

Product	Midea M-Thermal 30kW R290 commercial heat pump (OEM/white-label)
Pricing	Ex-factory China price approximately 1,800–2,500 USD; UK landed and certified price estimated 4,000–6,500 GBP depending on certification scope
Tech Specs	R290 refrigerant, 30kW output at A7/W45, EVI (Enhanced Vapour Injection) compressor for low ambient operation to -25°C, SCOP ~4.0, modbus/BACnet communications, stainless steel casing option, multiple unit cascade support

STRENGTHS

Very low manufacturing cost base, high production volume, broad OEM customisation options, improving quality consistency, increasingly seeking EU/UK market certification

WEAKNESSES

Brand perception gap in UK professional market, MCS certification not yet established for all models, after-sales support infrastructure in UK immature, installer confidence lower than European brands

DIFFERENTIATION ANGLE

A European-branded or UK-designed product at similar price using Midea-class manufacturing could bridge the trust gap, offering competitive pricing with Western brand credibility, certification, and local support that pure Chinese OEM cannot yet match

Vaillant

Product	aroTHERM plus VWL 125/5 AS (R290, scalable to 30kW with cascade)
Pricing	Single unit approximately 4,000–5,500 GBP trade; cascade system for 30kW total approximately 9,000–12,000 GBP installed hardware cost
Tech Specs	R290 refrigerant, single unit 12.5kW (cascade to 30kW+), SCOP 5.0 at A7/W35, operating range -20°C to +43°C ambient, ErP A+++ rated, integrated eBUS controller, quiet mode operation 40 dB(A), weight 170kg per unit

STRENGTHS

Excellent SCOP performance, strong UK installer network and training, MCS certified, reliable brand, good spare parts availability, modular cascade approach provides flexibility

WEAKNESSES

30kW in single package not available – requires cascade increasing installation complexity and cost, cascade controller adds cost and potential failure points, not optimised for single-unit 30kW commercial applications

DIFFERENTIATION ANGLE

A single-package 30kW R290 unit at 4,500 GBP eliminates the cascade complexity and cost premium of Vaillant's approach, offering installers a simpler, cheaper solution for 30kW projects without multi-unit pipework and commissioning overhead

Data Sources

[LLM Search] Aggregated competitor data via LLM

Overall Section Grade: D

1.4 Constraints & Targets ^[A]

Target Material	Not computed
Target Process	Not computed
Tolerance Target	Not computed
Quantity Target	Not computed
Batch Size	undefined
Target Cost Ceiling	Not computed
Max Mass	Not computed

Research Sources

Title	Type	Year	Relevance
UK Boiler Upgrade Scheme – GOV.UK	government_policy	2026	Defines the 7,500 GBP grant available for heat pump installations, directly impacting market demand and the competitive pricing landscape for 30kW units.
EU F-Gas Regulation (EU) 2024/573	regulation	2026	Outlines the phase-down schedule for high-GWP refrigerants, accelerating adoption of natural refrigerants like R290 across Europe and the UK.
EN 378:2016+A1:2020 – Refrigerating Systems and Heat Pumps: Safety and Environmental Requirements	technical_standard	2026	Primary safety standard governing design, manufacture, and installation of R290 heat pumps, with specific requirements for flammable refrigerant charge limits.
BSRIA World Market for Heat Pumps 2023	market_research	2026	Provides market sizing, growth forecasts, and competitive landscape for heat pumps in Europe, including natural refrigerant trends.
IEA Heat Pumps Technology Report 2023	industry_report	2026	Covers global heat pump deployment trends, technology developments in natural refrigerants, and policy drivers supporting R290 adoption.

MCS 020 Heat Pump Standard – Microgeneration Certification Scheme	certification_standard	2026	UK certification framework required for heat pumps to qualify for government incentives and installer compliance, critical for market access.
Stiebel Eltron WPL-A Product Range Technical Documentation	competitor_datasheet	2026	Reference for incumbent 30kW air source heat pump specifications, pricing benchmarks, and feature sets against which a new R290 unit must compete.

Data Sources

[Mixed] Constraints provided by User, Sources generated by LLM

Overall Section Grade: B

2.1 EN 378-1:2016+A1:2020 [D]

Refrigerating Systems and Heat Pumps – Safety and Environmental Requirements

Status: not-started

Owner: Lead Mechanical/Refrigeration Engineer

Summary

Comprehensive safety standard covering refrigerant classification, charge limits, system design, installation, and risk assessment for refrigerating systems including heat pumps using flammable refrigerants like R290.

Applicability

UK and EU markets; mandatory for CE/UKCA marking; applies to design, manufacture, and installation of the 30kW R290 unit

Engineering Impact

Dictates maximum allowable R290 charge based on installation category, requires leak detection systems if charge exceeds thresholds, mandates pressure relief, and influences enclosure ventilation design

EVIDENCE REQUIRED

Risk assessment documentation, refrigerant charge calculation report, pressure vessel test certificates, conformity declaration, technical file

GAP ACTION

Conduct preliminary charge size calculation for 30kW duty, identify applicable installation category, and commission EN 378 gap analysis with a notified body

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.2 IEC 60335-2-89 / BS EN 60335-2-89 [D]

Household and Similar Electrical Appliances – Safety: Particular Requirements for Commercial Refrigerating Appliances with an Incorporated or Remote Refrigerant Unit

Status: not-started

Owner: Electrical/Safety Engineer

Summary

Covers electrical safety requirements for refrigerating appliances including those using flammable refrigerants, with specific annexes (Annex GG, HH) addressing A3 refrigerants such as R290 at higher charge levels.

Applicability

Required for UKCA/CE marking; applies to electrical components, controls, compressor wiring, and flammable refrigerant risk mitigation within the product

Engineering Impact

Requires specific wiring harness protection, compressor motor enclosure ratings, spark-free electrical components in refrigerant zones, and software-controlled leak response

EVIDENCE REQUIRED

Electrical safety test reports from accredited laboratory, ATEX/IECEx component certificates for components in refrigerant-contact zones, software safety documentation

GAP ACTION

Identify all electrical components within or adjacent to the refrigerant circuit and verify compliance with Annex GG flammable refrigerant requirements; engage test house early

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.3 MCS 020 [D]

MCS Heat Pump Standard – Microgeneration Certification Scheme

Status: not-started

Owner: Product Compliance Manager

Summary

UK-specific product certification standard required for heat pumps to be eligible for the Boiler Upgrade Scheme (BUS) government grant. Covers performance, noise, and installation requirements.

Applicability

UK market only; essential for installer eligibility under BUS; strongly influences purchasing decisions by MCS-certified installers

Engineering Impact

Mandates declared COP/SCOP values, noise level limits (typically <42 dB(A) at 1m for residential proximity), and compliance with MCS Product Certification criteria including EN 14825 testing

EVIDENCE REQUIRED

EN 14825 seasonal performance test report from UKAS-accredited laboratory, noise test data, product data sheets in MCS format, third-party product certification from MCS-approved certification body

GAP ACTION

Register with an MCS-approved certification body (e.g., Kiwa, BRE), initiate EN 14825 performance testing, and prepare technical documentation file per MCS 020 requirements

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.4 UKCA / CE Marking (UK SI 2016/1101 & EU 2006/42/EC) ^[D]

Machinery Directive / UK Machinery Regulations – UKCA and CE Marking

Status: not-started

Owner: Regulatory/Compliance Director

Summary

Requires heat pump products to meet essential health and safety requirements before being placed on the UK (UKCA) and EU (CE) markets. Covers mechanical safety, electrical safety, and risk management documentation.

Applicability

Mandatory for sale in UK and EU; requires Declaration of Conformity and technical construction file; applies to the complete assembled heat pump unit

Engineering Impact

Requires formal risk assessment (ISO 12100), guarding of moving parts (fans, compressor), emergency stop provisions, and complete technical file including all sub-component certificates

EVIDENCE REQUIRED

Technical construction file, Declaration of Conformity (DoC), risk assessment per ISO 12100, harmonised standard test reports, instructions for use in English

GAP ACTION

Appoint UK Responsible Person and EU Authorised Representative, initiate technical file compilation, and map all applicable harmonised standards to the product design

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.5 ErP Directive / Ecodesign Regulation (EU) 2016/2281 & UK Equivalent ^[D]

Ecodesign for Energy-Related Products – Space Heaters and Combination Heaters

Status: not-started

Owner: Energy Performance Engineer

Summary

Sets minimum seasonal space heating energy efficiency requirements for heat pumps and mandates energy labelling. Requires minimum SCOP values and seasonal energy efficiency declarations.

Applicability

UK and EU markets; applies to heat pumps used for space heating output e10kW; enforced through market surveillance

Engineering Impact

Minimum SCOP of 3.10 (medium temperature) or 2.50 (high temperature) for air-source heat pumps; influences refrigerant circuit design, heat exchanger sizing, and inverter compressor selection to meet efficiency thresholds

EVIDENCE REQUIRED

EN 14825 seasonal performance test results, energy label artwork approved by certification body, product fiche data sheet, technical documentation file

GAP ACTION

Define target SCOP for design specification, model seasonal performance using EN 14825 climate bins, and validate against Ecodesign minimum thresholds before finalising refrigerant circuit design

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

3. Sizing & Spatial Optimisation ^[D]

Sizing data not computed.

5. Cost Waterfall & Economics [D]

Unit Cost Breakdown

Module	Subtotal
Total Unit BOM Cost	Not computed

Non-Recurring Engineering (NRE)

Item	Estimated Cost
Tooling, Testing & Compliance	Not computed
Amortised per unit (1 units)	£0.00

Cost Reduction Paths

Strategy	Est. Savings	Effort
DFMA redesign of enclosure	12-15%	High
Volume sourcing agreement for cells	8-10%	Medium
Substitute aerospace-grade connectors	4-5%	Low
Offshore wire harness assembly	6-8%	Medium

Data Sources

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

Overall Section Grade: C

6. Failure Modes & Effects Analysis (FMEA) [D]

Module	Hazard	Cause	Effect	S×L=RPN	Mitigation & Verification Test
No risk matrix data available.					

Data Sources

[LLM Synthesis] FMEA Matrix Generation

Overall Section Grade: E

7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

Section	Grade	Primary Source Type
1. Brief & Requirements	B	User Input & Verified Search
2. Regulatory Posture	D	LLM Knowledge Base
3. Sizing Optimization	B	Deterministic Solver
4. System Modules & Specs	D	Algorithmic Extraction + LLM
5. Economics Waterfall	C	Arithmetic Model + Estimations
6. FMEA Risk	E	Pure LLM Synthesis

8. Audit Log

Pipeline execution trace for traceability and verification.

#	Phase	Action	Result
1	Ingest	Parsed unstructured text brief	Extracted variables
2	Research	Scanned for market and competition	Built landscape model
3	Regulatory	Mapped regional standards	Identified 5-10 standard codes
4	Architecture	Decomposed to functional modules	0 modules created
5	Sizing	Algorithmic 3D packing	Constraints failed
6	BOM	Synthesised constituent parts	0 lines generated
7	Sourcing	Matched parts to supplier APIs	Selected preferred vendors
8	Economics	Applied cost models	Computed Not computed
9	Review	Cross-specialist critique	Addressed engineering conflicts
10	Risk	Populated FMEA table	Computed RPN values
11	Publish	Generated final PDF report	Success